

A One-Scalar Implicit Pac-Man Function

A short note on a max-based level-set representation

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Abstract

This note records a compact scalar-valued function whose zero set is a Pac-Man-shaped planar curve and whose nonpositive sublevel set is the corresponding filled region. The construction is a level-set description: it combines the unit disk and the complement of a symmetric wedge by taking the maximum of two scalar constraint functions. The geometry itself—a disk with a wedge removed—appears in published mathematical literature; the purpose here is to isolate a particularly concise one-scalar implicit representation and relate it to those sources. No claim of priority for the underlying Pac-Man geometry is intended.

1 Definition

For a half-angle

$$0 < \alpha < \frac{\pi}{2},$$

define the scalar function

$$F_\alpha(x, y) = \max(x^2 + y^2 - 1, (\tan \alpha)x - |y|). \quad (1)$$

The associated filled Pac-Man set is the nonpositive sublevel set

$$P_\alpha = \{(x, y) \in \mathbb{R}^2 : F_\alpha(x, y) \leq 0\}, \quad (2)$$

and its outline is the zero level set

$$\partial P_\alpha = \{(x, y) \in \mathbb{R}^2 : F_\alpha(x, y) = 0\}. \quad (3)$$

This is an *implicit* function description. It is not an ordinary graph of the form $y = f(x)$; rather, one scalar field on the Cartesian plane encodes the entire two-dimensional region and boundary.

The key identity is

$$\max(a, b) \leq 0 \iff a \leq 0 \text{ and } b \leq 0.$$

Applying it to (1) gives

$$P_\alpha = \{x^2 + y^2 \leq 1\} \cap \{|y| \geq (\tan \alpha)x\} \quad (4)$$

$$= \{x^2 + y^2 \leq 1\} \setminus \{x > 0, |y| < (\tan \alpha)x\}. \quad (5)$$

Thus the construction starts with the unit disk and removes a wedge pointing in the positive x direction. The wedge has full opening angle 2α .

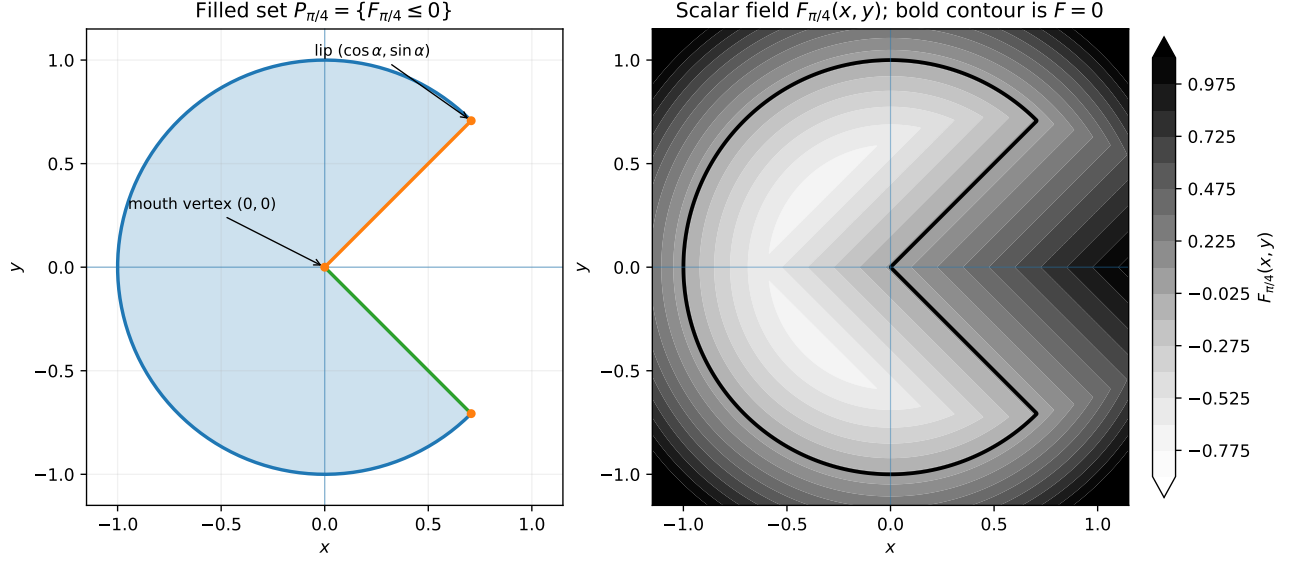


Figure 1: The construction for $\alpha = \pi/4$. Left: the filled set $P_{\pi/4}$. Right: the scalar field $F_{\pi/4}(x, y)$, with the zero contour emphasized.

2 Geometry and elementary properties

The two mouth edges are

$$y = \pm(\tan \alpha)x, \quad 0 \leq x \leq \cos \alpha,$$

and meet the unit circle at the two “lip” points

$$(\cos \alpha, \pm \sin \alpha).$$

Equivalently, the boundary consists of the circular arc

$$(\cos \theta, \sin \theta), \quad \alpha \leq \theta \leq 2\pi - \alpha,$$

together with the two unit radial segments from the origin to the lip points. The origin is the re-entrant mouth vertex.

Because the removed portion is a unit circular sector of angle 2α , the area and perimeter are particularly simple:

$$\boxed{\text{Area}(P_\alpha) = \pi - \alpha,} \quad \boxed{\text{Perimeter}(P_\alpha) = 2\pi - 2\alpha + 2.} \quad (6)$$

For the visually symmetric case $\alpha = \pi/4$, the mouth has a 90° opening and

$$F_{\pi/4}(x, y) = \max(x^2 + y^2 - 1, x - |y|),$$

with area $3\pi/4$ and perimeter $3\pi/2 + 2$.

The function F_α is continuous and locally Lipschitz. It is piecewise smooth, but it is not differentiable everywhere: the absolute value creates a kink, and the max operation is nonsmooth where its two arguments exchange dominance. These nonsmooth features are natural for a shape whose mouth contains corners. If a smooth approximation is needed for optimization or numerical work, the absolute value and maximum can be regularized, although an approximation generally changes the exact zero set slightly.

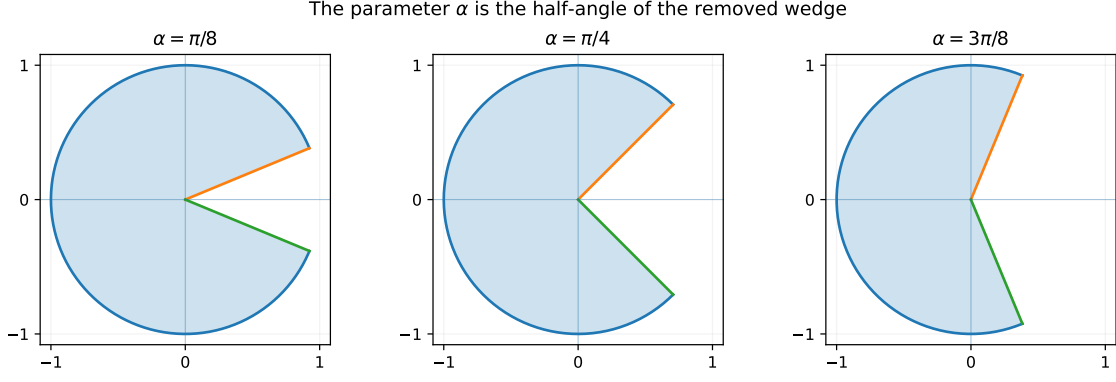


Figure 2: The same scalar construction for three half-angles. Increasing α widens the mouth; the full wedge opening is 2α .

A translated, rotated, radius- R version can be written without changing the idea. If (c_x, c_y) is the center and ϕ is the mouth direction, set

$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} x - c_x \\ y - c_y \end{pmatrix}.$$

Then the dimensionless scalar field

$$F_{\alpha, R, c, \phi}(x, y) = \max\left(\frac{u^2 + v^2}{R^2} - 1, \frac{(\tan \alpha)u - |v|}{R}\right)$$

encodes a radius- R Pac-Man centered at $c = (c_x, c_y)$ and facing angle ϕ .

3 Relation to existing literature

The compact formula (1) sits at the intersection of several established ideas: Pac-Man-shaped domains, level-set descriptions of geometry, oriented or approximate distance functions, and R-function/constructive-solid-geometry combinations. The literature below is ordered from the closest geometric precedents to broader implicit-function frameworks.

Caboussat, Glowinski, and Gourzoulidis (2022). In numerical experiments for a nonsmooth prescribed Jacobian equation, the authors define the unit disk Ω_d and a “pacman” domain

$$\Omega_p = \Omega_d \setminus \{(x_1, x_2) : x_1 > 0, |x_2| < x_1\}.$$

This is the same geometry as $P_{\pi/4}$ in (5), up to open/closed-boundary conventions. Their paper expresses the object as a set difference rather than as the zero/sublevel set of the single scalar function $\max(x^2 + y^2 - 1, x - |y|)$ [1].

Levenberg and Ma’u (2020). Their paper *Monge–Ampère of Pac-Man* studies a nonconvex Pac-Man set described as a disk minus a centrally symmetric wedge. Their principal example has a wedge opening of $\pi/2$ and a mouth vertex at the disk center. After translation of coordinates, this is again the same basic geometry as the $\alpha = \pi/4$ case here; their angle parameter denotes the full wedge opening, whereas α in this note is the half-angle [2].

Beneš (2021). Beneš studies planar “Pacman domains” of the form

$$D_\beta(n) = \{re^{i\theta} : 0 < \theta < 2\pi - \beta, 0 < r < 2n\} - z_0,$$

for estimates on discrete Green’s functions. This is another disk-with-wedge-removed family, written in polar/complex notation and at a different scale and translation [3].

Jankowski and Stanberry (2012). In work on oriented distance functions for random sets, the authors give an explicit example called “pacman in $\mathbb{R}^2$ ”: a disk centered at the origin with one quadrant removed. They then study the Pac-Man through its *oriented distance function* (ODF), so the shape is encoded by a single scalar field whose zero set is the boundary. The particular mouth orientation and angle differ from (1), and the ODF is not the max expression used here, but this is a direct precedent for treating a two-dimensional Pac-Man as a scalar implicit object rather than only as a drawn set [5].

Mulroy, Lopez, Spenko, and Srivastava (2022). These authors use R-functions and approximate distance fields to control the shape of soft robots. One of their explicit demonstrations is a Pac-Man-like shape. They state that the distance function for that example is constructed by R-equivalence of two line segments and an arc, yielding a single analytical scalar field for the desired outline. This is very close in spirit to the present note: simple boundary primitives are combined into one implicit function, although their smooth approximate-distance construction is different from the nonsmooth max/absolute-value formula (1) [6].

Schaffer and Exl (2024). In a constructive-solid-geometry framework based on R-functions, Schaffer and Exl explicitly discuss Boolean combinations of implicit geometries and include min/max systems among the R-function choices. Their code example constructs a three-dimensional “Pacman” by subtracting a mouth from a sphere, and they plot the zero level set of the resulting approximate distance function. Thus the paper contains the same broad recipe—primitive implicit functions, Boolean-style scalar composition, and a Pac-Man zero level set—though in three dimensions and with a different R-function formula [4].

Neiva and Badia (2021). In their work on h -adaptive aggregated unfitted finite elements, the authors use several embedded interfaces represented as the zero level set of a (piecewise-)smooth scalar function. Their two-dimensional test geometries explicitly include a “pacman” shape, which is also used for a Fichera-corner benchmark. The paper therefore supplies another peer-reviewed instance of

$$\text{Pac-Man boundary} = \{(x, y) : \phi(x, y) = 0\},$$

although it refers elsewhere for the detailed level-set formula and does not print the compact max expression (1) [7].

Burman, Claus, Hansbo, Larson, and Massing (2015). The CutFEM framework represents embedded geometry by level-set functions on a background mesh. As a basic example, the paper uses

$$\phi(x, y) = x^2 + y^2 - 1$$

for a circular domain. This is exactly the first primitive appearing in (1). CutFEM is not itself a Pac-Man construction, but it provides a standard numerical-analysis setting in which a scalar function’s sign encodes the inside, outside, and boundary of a planar region [8].

Earlier R-function background. Gladkov and Svendsen present the R-functions method for solving elliptic PDEs on complex two-dimensional domains, illustrating the longer mathematical tradition of building exact or approximate analytical domain descriptions from simpler implicit pieces [9]. This paper is background rather than an exact Pac-Man match, but it helps place the later Pac-Man R-function examples in context.

Smooth polar Pac-Man curves. A non-peer-reviewed Mathematics Stack Exchange discussion asks for smooth polar equations $r = r(\theta)$ that trace a Pac-Man-like outline. Those constructions are conceptually different: they parameterize a curve radially, while (1) represents the filled region and its boundary as sublevel and zero sets of a scalar field on $\mathbb{R}^2$ [10].

The absolute-value term in (1) can itself be rewritten as a min of two half-plane functions,

$$(\tan \alpha)x - |y| = \min((\tan \alpha)x - y, (\tan \alpha)x + y).$$

Consequently,

$$F_\alpha(x, y) = \max(x^2 + y^2 - 1, \min((\tan \alpha)x - y, (\tan \alpha)x + y)),$$

which makes the constructive-solid-geometry connection especially transparent: a circular implicit primitive is combined with two linear half-plane primitives using min/max operations.

Across the sources located for this note, the disk-minus-wedge Pac-Man geometry is established, Pac-Man zero level sets and distance functions are established, and min/max or R-function composition of implicit primitives is established. What was *not* located in the sources searched is the particular compact one-parameter family

$$F_\alpha(x, y) = \max(x^2 + y^2 - 1, (\tan \alpha)x - |y|)$$

written in this exact or an algebraically identical form and presented specifically as a Pac-Man function. This statement is intentionally limited: it records the outcome of the literature search summarized here and is not a comprehensive claim of mathematical priority.

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